CSV102:ADVANCE LINUX

L:3 T:0 P:0 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: understand advanced Linux system administration and configuration techniques

CO2 :: understand advanced networking and security configurations for cloud environments

CO3 :: analyze cloud-based tools and services for managing and deploying Linux-based applications

CO4 :: understand troubleshooting and optimization Linux systems for performance and efficiency

CO5 :: analyze advanced Linux skills to real-world cloud computing scenarios

CO6 :: understand Linux monitoring and adminsitration

Unit I

Advanced Linux System Administration : Advanced file system management and troubleshooting, User and group management with advanced techniques, Process management and scheduling for optimal performance, Resource allocation and monitoring for cloud-based applications

Unit II

Advanced Networking and Security for Cloud Environments : Advanced network configuration and troubleshooting for cloud environments, Implementing firewalls and network security measures to protect cloud-based systems, Configuring network services(DHCP, DNS,VPN) for cloud-based infrastructure, Understanding and mitigating cloud-based security threats

Unit III

Cloud-Based Linux Application Management : Containerization technologies (Docker,Kubernetes) for scalable and manageable cloud-based applications, Utilizing cloud-based tools (AWSEC2,Azure Virtual Machines)for deploying and managing Linux instances, Implementing cloud-based configuration management tools(Ansible,Chef) for automated provisioning and configuration, Packaging and deploying Linux applications using cloud-based package managers(RPM,APT)

Unit IV

Linux Performance Optimization and Troubleshooting : Performance tuning and optimization techniques for cloud-based Linux systems, Identifying and resolving performance bottlenecks in cloud-based applications, Utilizing system monitoring tools (ps,top,htop) for analyzing system performance, Troubleshooting common Linux system issues and errors

Unit V

Advanced Linux for Cloud Computing Scenarios : Designing and implementing cloud-based Linux infrastructure for various scenarios, Automating cloud-based Linux tasks using scripting and configuration management tools, Integrating cloud-based Linux systems with cloud-native services (AWS Lambda, Azure Functions), Managing and maintaining cloud-based Linux systems for long-term reliability and security, Understand how to install, update, and remove packages using APT tools

Unit VI

Linux Monitoring : Introduction to Monitoring in Linux, Linux Monitoring Metrics, Linux In-built Performance Monitoring Tools, Other Monitoring Tools, Linux Monitoring using SNMP, Third-Party Monitoring Tools

Text Books:

1. LINUX POCKET GUIDE by DANIEL J. BARRETT, O'reilly Media

References:

1. RUNNING LINUX by MATTHIAS KALLE DALHEIMER, LAR KAUFMAN, MATT WELSH, O'REILLY

2. HOW LINUX WORKS by BRIAN WARD, NO STARCH PRESS

3. LINUX SYSTEM ADMINISTRATION: SOLVE REAL-LIFE LINUX PROBLEMS QUICKLY by TOM ADELSTEIN, BILL LUBANOVIC, O'REILLY

4. LINUX FOR BEGINNERS by JASON CANNON, independent publisher

5. LINUX COMMAND LINE AND SHELL SCRIPTING BIBLE by RICHARD BLUM, WILEY